

EP02 — What is an algorithm?

ENGLISH

01 Title

What is an **algorithm**? It sounds technical and scary — but here's a secret: you already use algorithms every single day, without even noticing. Let's break it down.

[CUE] Hold the title ~2s. Friendly, confident — "this is easier than you think".

02 The Hook

Did you make 2-minute Maggi today? Then you ran an algorithm. Boil the water, add the masala, add the Maggi, wait two minutes. **Steps, in a fixed order**, to get a result. That is **exactly** what an algorithm is — a recipe.

[CUE] Energy up. Count the steps on your fingers as you say them.

03 Today's question

So today's question: what **exactly** is an algorithm — and why does computer science care about it so much? Let's find out.

[CUE] Slow down. Let the question sit for a beat.

04 Idea 1 — A precise recipe

Idea one. An algorithm is just a **step-by-step recipe** to solve one specific problem. Look at making chai: boil water, add tea and sugar, add milk, boil two minutes, strain. A clear set of steps, in order, that gets the job done. That's all an algorithm is.

[CUE] Point at the recipe block; read the steps with rhythm.

05 Idea 2 — Clear, ordered, finite

Idea two — but the steps need three rules. They must be **clear**: no “add a little” — how much, exactly? They must be **ordered**: add the Maggi before boiling the water, and it’s a disaster — order matters. And they must be **finite**: the recipe has to end. A computer follows your steps **literally**, so if you’re vague, it breaks.

[CUE] Stress clear / ordered / finite as three distinct beats.

06 Idea 3 — Efficiency

Idea three — and this is why CS is obsessed with algorithms. The **same task** can be solved fast or slow. Say you’re finding a word in a 1000-page dictionary. Go page by page, and that’s up to a thousand checks. But open the middle, see if your word is before or after, and jump again — about ten checks. Both are correct. One is a **hundred times faster**.

[CUE] Reveal the two bars (1000 vs ~10) as you compare.

07 Idea 4 — The queue (FIFO)

Idea four — computers also organise **work** with simple rules. Think of a samosa line: whoever comes first, gets served first. That’s a **queue** — in CS we call it **FIFO**, First In, First Out. Print jobs, ticket counters, even your WhatsApp messages — they’re all handled with a queue.

[CUE] Trace the samosa line left-to-right: “in... in... in... out”.

08 Why it matters

And this is everywhere. Google Maps uses an algorithm to find your shortest route home. YouTube uses one to pick your next video. UPI uses one to verify a payment in two seconds. All of it is just clever step-by-step algorithms, running millions of times a second.

[CUE] List the three examples briskly; they’re familiar, let them land.

09 Common myth

Now the myth. People think an algorithm is scary maths that only coders and AI use. Not true. An algorithm is just **clear steps** to do a task — and you use them every day. Writing code is simply putting those steps down **precisely** enough for a computer to follow.

[CUE] Reveal the green "Reality" card after a beat on the myth.

10 Recap

Quick recap. One: an algorithm is **precise, ordered, finite steps** to solve a problem. Two: the same task can have a fast or a slow algorithm — and choosing the fast one is the heart of computer science.

[CUE] Crisp. These are the two lines to remember.

11 CTA

So next time someone says "algorithm", just think: recipe. Simple. If this made it click, subscribe to Katixo KhojLab. Next question: how does the internet actually reach your phone? See you there.

[CUE] Warm, friendly. Smile in your voice on the subscribe line.
